

Geodesics of Attention

Cinema, Gesture, and the Geometry of Perception

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Abstract

We propose a unified geometric framework in which cinema, painting, drawing, swipe-gesture input, and visual perception are instances of a single underlying structure: directed motion through a structured spatial or perceptual field. A film is modeled as a continuous path through a *scenario manifold* $\mathcal{S} = \mathcal{C} \times \mathcal{A} \times \mathcal{L} \times \mathcal{E}$ whose factors encode camera configuration, actor pose, lighting, and environment. The perceptual impact of that path is determined not by its geometry in $\mathcal{S}$ but by its image under a rendering-and-perception map $\Pi : \mathcal{S} \rightarrow \mathcal{P}$ into a *perceptual manifold* $\mathcal{P}$ whose coordinates are attentional focus, revealed narrative information, occlusion structure, emotional intensity, and compositional balance.

Our central result—the *Geodesic Principle of Cinematic Naturalness*—states that cinematographically natural camera movements correspond to paths that locally minimize the perceptual action functional on $\mathcal{P}$, and that the metric governing this functional is the Fisher-Rao metric on the statistical manifold of viewer attentional distributions. Narrative intent is encoded as a potential function that forces paths toward perceptually salient states, yielding a variational equation that unifies classical editing rules (the 180-degree rule, match cuts, continuity editing, montage) as geometric constraints in $\mathcal{P}$.

We then demonstrate that the same variational structure governs painting strokes guided by image-gradient fields, children’s drawing as a sequential grammar of spatial action, QWERTY swipe gestures as a trajectory language over a discrete keyboard manifold, and eye saccades as perceptual sampling paths. The topology of gesture classes provides additional invariants that explain why swipe keyboards tolerate large motor variability. We further argue that artistic development constitutes a progressive shift from object-graph representations to field-based and finally radiance-model representations of the canvas. The framework connects cinematography, game-engine simulation, information geometry, embodied cognition, and trajectory-based generative models.

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1 Introduction

Many cultural activities that seem superficially unrelated share a common deep structure: they are all instances of directed motion through a structured spatial or perceptual field, where meaning or aesthetic quality emerges from the shape of that motion rather than from its endpoints alone. A filmmaker choosing a camera angle, a painter laying down a brushstroke, a child completing a figure on a page, a typist swiping a word, and a viewer’s eye jumping between salient features of a scene are all performing versions of the same cognitive and bodily operation.

This paper proposes a unified mathematical framework for these activities. The central object is a *trajectory*: a continuous path $\gamma(t)$ through some structured space, where the space is endowed with a geometry that determines which paths are natural, smooth, or costly. We argue that the appropriate notion of naturalness is always determined by a *perceptual metric*—a metric tensor on a manifold of experiential states—rather than by a purely physical or symbolic metric.

The framework has five main components.

- (1) **Scenario manifold $\mathcal{S}$** : the product of camera, actor, lighting, and environment configuration spaces.
- (2) **Perceptual manifold $\mathcal{P}$** : the space of viewer experiential states, carrying a metric derived from information geometry.
- (3) **Rendering-and-perception map $\Pi : \mathcal{S} \rightarrow \mathcal{P}$** : the operator that converts scenario states into perceptual states.
- (4) **Geodesic principle**: cinematographic naturalness corresponds to approximate geodesic motion in $\mathcal{P}$ subject to a narrative potential function.
- (5) **Generality**: the same variational structure governs painting strokes, drawing grammars, swipe gestures, eye saccades, and musical perception.

Modern film production already instantiates part of this structure. Virtual production systems such as Unreal Engine represent a scene as a game state $g = (C, A, L, E)$ whose components are camera pose, actor configurations, lighting, and environment, and the camera path through that state space is literally a trajectory. But a film recorded with a physical camera also constitutes a path through a latent scenario space, whether or not that space is ever explicitly simulated. The theoretical reinterpretation proposed here treats this implicit structure as the primary object of analysis.

Section 2 defines scenario space formally. Section 3 introduces the perceptual manifold and its metric. Section 4 states the central theorem and derives the variational equations. Section 5 reinterprets classical editing conventions. Section 6

grounds the metric in information geometry. Section 7 extends the framework to painting and visual gradients. Section 8 addresses children’s drawing and the grammar of spatial action. Section 9 treats QWERTY swipe gestures as a trajectory language. Section 10 develops topological invariants of gesture classes. Section 11 connects the framework to eye saccades and embodied cognition. Section 12 makes explicit the equivalence between painting traces and other trajectory systems. Section 13 analyzes curvature control in cinematic motion. Section 14 describes the developmental arc from object grammar to radiance model. Section 15 discusses implications for generative models. Section 16 concludes.

2 Scenario Space

2.1 Product structure

The scenario manifold is the product

$$\mathcal{S} = \mathcal{C} \times \mathcal{A} \times \mathcal{L} \times \mathcal{E},$$

where $\mathcal{C}$ is the camera configuration space, $\mathcal{A}$ is the actor configuration space, $\mathcal{L}$ is the lighting space, and $\mathcal{E}$ is the environmental state space. Each factor is a smooth manifold, so $\mathcal{S}$ inherits a product smooth structure.

Definition 2.1 (Scenario state). A *scenario state* is a tuple

$$s = (C, A, L, E) \in \mathcal{S}$$

where:

- $C = (x, y, z, \theta, \phi, f) \in \mathcal{C} \cong \mathbb{R}^3 \times S^2 \times \mathbb{R}_{>0}$ encodes camera position, orientation, and focal length;
- $A = (a_1, \dots, a_k) \in \mathcal{A}$ encodes the pose and animation state of each of k actors, with each a_i in a body-pose Lie group;
- $L \in \mathcal{L}$ encodes lighting configuration (source positions, intensities, color temperatures);
- $E \in \mathcal{E}$ encodes environmental parameters (weather, time of day, set geometry).

For discrete categorical parameters (shot type, costume change) the smooth structure is replaced by a stratified space. We work in the smooth case throughout.

2.2 Films as paths

Definition 2.2 (Film). A *film* is a piecewise-smooth path

$$\gamma : [0, T] \rightarrow \mathcal{S}.$$

Each frame corresponds to a sample $s_t = \gamma(t)$. The observable image at time t is the output of a rendering operator:

$$I_t = R(s_t), \quad R : \mathcal{S} \rightarrow L^2(\Omega, \mathbb{R}^3),$$

where $\Omega \subset \mathbb{R}^2$ is the image plane and $L^2(\Omega, \mathbb{R}^3)$ is the space of square-integrable color-valued functions on Ω . The observable film is the family $(I_t)_{t \in [0, T]}$, i.e., the image of the trajectory under R .

Remark 2.3. Writing $R : \mathcal{S} \rightarrow L^2(\Omega, \mathbb{R}^3)$ makes explicit that an image is an element of an infinite-dimensional function space, not merely a finite array. In a real-time game engine, $\mathcal{S}$ is the engine’s world state and R is the rendering pipeline. A film recorded on a physical set has the same abstract structure, with R replaced by the physical optics of the camera and the photochemical or electronic response of the sensor.

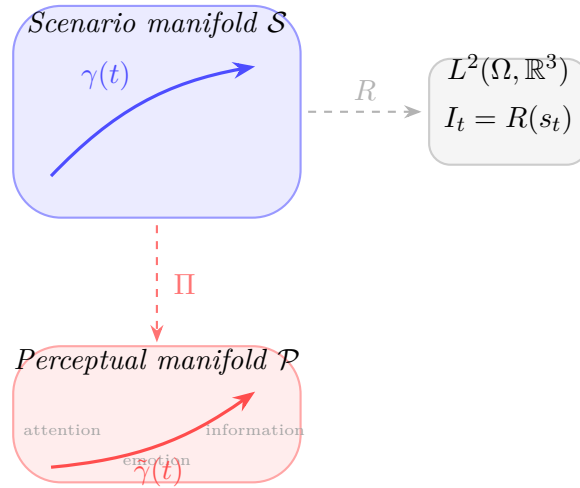


Figure 1: The scenario manifold $\mathcal{S}$, rendering map R , and perceptual manifold $\mathcal{P}$. A film is a path $\gamma(t)$ in $\mathcal{S}$; its aesthetic quality is determined by the induced perceptual path $\tilde{\gamma}(t)$. Selected coordinate axes of $\mathcal{P}$ are indicated inside the lower blob.

3 The Perceptual Manifold

The scenario manifold describes what the world *is*; the perceptual manifold describes what a viewer *experiences*. A physically simple camera move can feel disorienting, while a physically elaborate tracking shot can feel effortless. The difference lies not in $\mathcal{S}$ but in $\mathcal{P}$.

3.1 Perceptual coordinates

Definition 3.1 (Perceptual state). A *perceptual state* is a point

$$p = (a, r, o, e, q) \in \mathcal{P} \subset \Delta(\mathcal{X}) \times [0, 1]^4$$

where $\Delta(\mathcal{X})$ is the probability simplex over the set $\mathcal{X}$ of salient scene objects, and:

- $a \in \Delta(\mathcal{X})$ is the viewer's attentional distribution over salient objects;
- $r \in [0, 1]$ is revealed narrative information (cumulative proportion of withheld story information disclosed);
- $o \in [0, 1]$ is occlusion coherence (stability of depth ordering and figure-ground relations);
- $e \in [0, 1]$ is emotional intensity (arousal level induced by the scene);
- $q \in [0, 1]$ is compositional balance (degree of visual equilibrium in the frame).

Treating a as a probability distribution rather than a scalar makes the embedding $\mathcal{P} \subset \Delta(\mathcal{X}) \times [0, 1]^4$ natural and connects immediately to the Fisher-Rao metric developed in Section 6.

3.2 Riemannian metric and perception map

The perceptual manifold $\mathcal{P}$ carries a Riemannian metric tensor $g_{ij}(p)$ that measures the cognitive cost of moving between perceptual states. The infinitesimal perceptual distance is

$$ds^2 = \sum_{i,j} g_{ij}(p) dp^i dp^j.$$

Definition 3.2 (Rendering-and-perception map). The *rendering-and-perception map* is a smooth map

$$\Pi : \mathcal{S} \rightarrow \mathcal{P}$$

that assigns to each scenario state the perceptual state it induces in a standard viewer. A film trajectory $\gamma(t)$ in $\mathcal{S}$ induces the *perceptual trajectory*

$$\tilde{\gamma}(t) = \Pi(\gamma(t)) \in \mathcal{P}.$$

The metric on $\mathcal{P}$ pulls back to a metric on $\mathcal{S}$ via the Jacobian $D\Pi_s : T_s\mathcal{S} \rightarrow T_{\Pi(s)}\mathcal{P}$:

$$(D\Pi^*g)_s(u, v) = g_{\Pi(s)}(D\Pi_s \cdot u, D\Pi_s \cdot v), \quad u, v \in T_s\mathcal{S}.$$

In this pullback metric, scenario directions that induce large perceptual change are far apart even if they are physically close, giving a precise sense in which Π compresses or stretches directions according to their perceptual significance.

4 Naturalness as a Geodesic Principle

4.1 Central theorem

Theorem 4.1 (Geodesic Principle of Cinematic Naturalness). *Let $\gamma : [0, T] \rightarrow \mathcal{S}$ be a scenario trajectory and $\tilde{\gamma} = \Pi \circ \gamma$ the induced perceptual trajectory in $(\mathcal{P}, g)$. Among all scenario trajectories connecting fixed perceptual endpoints $\tilde{\gamma}(0)$ and $\tilde{\gamma}(T)$, the cinematographically natural trajectories are those whose induced perceptual paths locally minimize the perceptual action*

$$\mathcal{A}[\tilde{\gamma}] = \frac{1}{2} \int_0^T g_{ij}(\tilde{\gamma}(t)) \dot{\tilde{\gamma}}^i(t) \dot{\tilde{\gamma}}^j(t) dt.$$

These critical paths satisfy the geodesic equation

$$\ddot{\tilde{\gamma}}^k + \Gamma_{ij}^k \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j = 0,$$

where Γ_{ij}^k are the Christoffel symbols of g .

Proof. The action $\mathcal{A}$ is the standard energy functional on a Riemannian manifold. Its Euler-Lagrange equations are derived from the Lagrangian $\mathcal{L}(p, \dot{p}) = \frac{1}{2} g_{ij}(p) \dot{p}^i \dot{p}^j$. Varying $\mathcal{A}$ with fixed endpoints and applying the Leibniz rule yields

$$\frac{d}{dt}(g_{k\ell} \dot{\tilde{\gamma}}^\ell) = \frac{1}{2}(\partial_k g_{ij}) \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j,$$

which, after raising an index with $g^{k\ell}$ and symmetrizing, gives precisely the geodesic equation with the standard Christoffel symbols $\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$ [7]. $\square$

The geodesic equation says that a naturally filmed sequence has no *perceptual acceleration*: the viewer's experiential state changes at a steady rate with no sudden jolts. This is the mathematical content of the intuitive principle that good cinematography feels effortless.

4.2 Narrative forcing

A purely geodesic path in $\mathcal{P}$ does not capture the director's intention to reveal information or guide attention toward specific story events. We introduce a *narrative potential* $V : \mathcal{P} \rightarrow \mathbb{R}$, where low values of V correspond to perceptually important or informationally rich states.

Definition 4.2 (Cinematic action with narrative potential). The *cinematic action* is

$$\mathcal{J}[\tilde{\gamma}] = \int_0^T \left[\frac{1}{2} g_{ij}(\tilde{\gamma}) \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j + \lambda V(\tilde{\gamma}) \right] dt,$$

where $\lambda > 0$ balances perceptual smoothness against narrative urgency.

Proposition 4.3. *The Euler-Lagrange equations of $\mathcal{J}$ are the narratively forced geodesic equation*

$$\ddot{\tilde{\gamma}}^k + \Gamma_{ij}^k \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j + \lambda g^{k\ell} \partial_\ell V(\tilde{\gamma}) = 0,$$

or equivalently, using the covariant derivative along the path,

$$\frac{D\dot{\tilde{\gamma}}}{dt} = -\lambda \nabla_g V(\tilde{\gamma}).$$

Remark 4.4. The parameter λ encodes directorial style. A director who prizes compositional fluidity operates at small λ (Tarkovsky, Ophüls); a director who prioritizes narrative information density operates at large λ (Eisenstein, Greengrass).

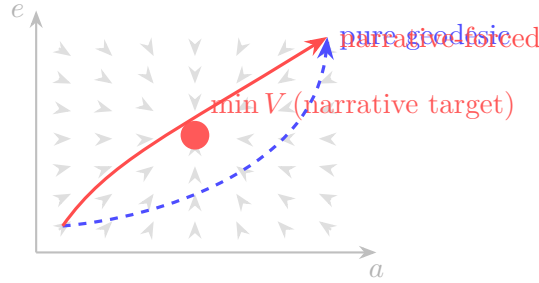


Figure 2: Narrative forcing bends the perceptual path toward the attractor $\min V$ (the narratively important state) at the cost of some smoothness. Coordinate axes a (the narratively important state) and e (emotional intensity) label two dimensions of $\mathcal{P}$.

5 Editing as Trajectory Transformation

Film editing introduces discontinuities or structured transformations in the scenario trajectory. We interpret classical editing operations as geometric operations on paths in $\mathcal{P}$, drawing on the theoretical framework of Bordwell and Thompson [4] for the underlying film-language taxonomy.

5.1 Cuts and perceptual distance

A *hard cut* at time t_0 is a jump discontinuity: $\gamma(t_0^-) \neq \gamma(t_0^+)$. In $\mathcal{P}$ the cost of the cut is proportional to $d_{\mathcal{P}}(\tilde{\gamma}(t_0^-), \tilde{\gamma}(t_0^+))$. A cut is *invisible* when this distance is small, i.e., when the cut preserves attentional focus a , occlusion coherence o , and compositional balance q . This is the geometric content of continuity editing.

5.2 Dissolves and wipes

A *dissolve* blends two scenario states:

$$I_t = (1 - \alpha(t)) R(\gamma_1(t)) + \alpha(t) R(\gamma_2(t)), \quad \alpha : [0, 1] \rightarrow [0, 1].$$

The dissolve is perceptually smooth when the source paths $\Pi(\gamma_1)$ and $\Pi(\gamma_2)$ are close in $\mathcal{P}$.

5.3 Montage

A *montage sequence* is a piecewise trajectory $\gamma = (\gamma_1, \gamma_2, \dots, \gamma_k)$ with jump discontinuities at the joints. Montage deliberately creates large perceptual jumps, but ideally along semantically meaningful axes (primarily e and r), so that perceptual cost is compensated by conceptual gain. Eisenstein’s principle of intellectual montage is precisely that large excursions in $\mathcal{P}$ along the r -axis (narrative revelation) can be aesthetically productive even at high perceptual cost in the a and o coordinates.

5.4 The 180-degree rule as a chart condition

Proposition 5.1. *The 180-degree rule is the condition that the camera trajectory remains within a single local chart of $\mathcal{P}$ with respect to the occlusion coordinate o . Crossing the line forces a coordinate singularity in that chart.*

Proof. If actors lie along an axis $\ell \subset \mathbb{R}^3$, a camera on one side of ℓ assigns consistent left-right screen positions to the actors. Crossing ℓ reverses this assignment, producing an abrupt change in o that cannot be connected by a path of small g -length in $\mathcal{P}$. Formally, the occlusion coordinate o has a fold singularity at the axial plane, so the camera trajectory must remain in one connected component of $\{o > 0\}$ to remain within a single chart. $\square$

5.5 Match cuts and eyeline matches

A *match cut* at time t_0 satisfies $\|\tilde{\gamma}_1(t_0^-) - \tilde{\gamma}_2(t_0^+)\|_{\mathcal{P}} < \varepsilon$ for small ε . The match cut is the editorial operation that minimizes perceptual distance across a cut. An *eyeline match* aligns the attentional distribution a of one shot with the implied object of attention in the next, preserving continuity in the $\Delta(\mathcal{X})$ component of $\mathcal{P}$.

6 Information Geometry of the Perceptual Metric

6.1 Viewer attention as a statistical manifold

We ground the metric g on $\mathcal{P}$ in information geometry by modeling the viewer’s attentional coordinate a as a probability distribution $P_t \in \Delta(\mathcal{X})$ over salient scene objects. When P_t is parameterized by $\theta \in \Theta$, the *Fisher information metric* is

$$g_{ij}^F(\theta) = \mathbb{E}_{P(\cdot; \theta)} \left[\frac{\partial \log P(x; \theta)}{\partial \theta^i} \frac{\partial \log P(x; \theta)}{\partial \theta^j} \right].$$

This is the unique (up to scale) Riemannian metric on $\Delta(\mathcal{X})$ invariant under sufficient statistics [5, 1].

Theorem 6.1. *The perceptual metric g on $\mathcal{P}$ can be taken to dominate the Fisher-Rao metric on the attentional component $\Delta(\mathcal{X})$. A scenario trajectory γ is perceptually smooth if and only if the induced attentional path $t \mapsto P_t$ has bounded Fisher-Rao velocity.*

Proof sketch. The attentional distribution P_t is the marginal of the full perceptual state over the a -component. Bounded Fisher-Rao velocity means $\int_0^T g_{ij}^F(P_t) \dot{P}_t^i \dot{P}_t^j dt < \infty$, which bounds the rate of change of all attentional statistics. By smoothness of the inclusion $\Delta(\mathcal{X}) \hookrightarrow \mathcal{P}$, this bounds the full perceptual action $\mathcal{A}[\gamma]$. $\square$

6.2 Continuity editing as optimal attentional transport

The cost of a cut from distribution P^- to P^+ can be measured by the Wasserstein-2 distance:

$$W_2(P^-, P^+) = \inf_{\pi \in \Pi(P^-, P^+)} \left(\int_{\mathcal{X}^2} d(x, y)^2 d\pi(x, y) \right)^{1/2},$$

where the infimum is over all couplings π with marginals P^- and P^+ [15]. A cut is invisible when $W_2(P^-, P^+)$ is small: viewer attention can be transported cheaply from the pre-cut to the post-cut distribution.

Corollary 6.2. *Continuity editing minimizes the expected Wasserstein-2 attentional transport cost across shot boundaries. Invisible cuts are precisely those for which the optimal transport plan is nearly the identity.*

This gives a precise measure-theoretic account of why continuity editing works, independent of any appeal to viewer habit or cultural convention.

7 Painting as Gradient-Field Integration

7.1 The canvas as an evolving scalar field

A painting-in-progress is a scalar field $B_t : \Omega \rightarrow \mathbb{R}$, where $\Omega \subset \mathbb{R}^2$ is the canvas. A brushstroke $\gamma_{\text{paint}} : [0, \tau] \rightarrow \Omega$ updates the canvas:

$$B_{t+dt}(x, y) = B_t(x, y) + \int_0^{dt} K(x - \gamma_x(t), y - \gamma_y(t)) dt,$$

where K is the brush footprint kernel. Painting is the process of *writing trajectories into a field*; crucially, each stroke modifies the field through which subsequent strokes move.

7.2 Gradient guidance

The structural information of the scene is encoded in the gradient $\nabla I = (I_x, I_y)$ of the luminance field $I : \Omega \rightarrow \mathbb{R}$, approximated in practice by Sobel filters: $G_x = S_x * I$, $G_y = S_y * I$, $|\nabla I| = \sqrt{G_x^2 + G_y^2}$. Painters often align strokes with the *isophote direction* $T = (-I_y, I_x)$, tangent to level curves of I , yielding the stroke differential equation

$$\frac{d\gamma_{\text{paint}}}{dt} = \alpha T(\gamma_{\text{paint}}(t)) + \beta \nabla I(\gamma_{\text{paint}}(t)).$$

This has the same form as the narratively forced geodesic of Section 4: motion follows a vector field derived from perceptual structure, balancing contour-following (α) against gradient-crossing (β) in a manner precisely analogous to the balance between perceptual smoothness and narrative urgency in cinematography.

7.3 Structure tensor and dominant orientation

The *structure tensor*

$$J = \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}$$

summarizes local orientation. Its dominant eigenvector gives the preferred direction of edges and forms [13, 9].

Proposition 7.1. *A painter who aligns brushstrokes with the dominant eigenvector field of J is integrating a trajectory through the metric induced by J on Ω . Each stroke is a geodesic segment in this derived metric.*

Painterly rendering algorithms such as those of Hertzmann [9] make this explicit computationally, generating curved brushstrokes aligned with image curvature flow—a direct algorithmic implementation of the gradient-field integration described here.

8 Drawing as a Sequential Grammar of Spatial Action

8.1 Goodnow’s grammar of action

Jacqueline Goodnow’s empirical studies demonstrate that children’s drawings are not arbitrary marks but records of structured action sequences [8]. Children follow systematic strategies for initiating figures, attaching new elements, and expanding outward—a procedural *grammar of action* governing how gestures combine to produce spatial structures.

In the present framework, a drawing is a sequence of trajectories $\gamma_1, \gamma_2, \dots, \gamma_n$ each depositing a stroke on the canvas field B , governed by an iterative mapping

$$B_{t+1} = F(B_t, a_t),$$

where a_t is the stroke action at step t . Goodnow’s attachment and expansion rules constrain F : new strokes are placed near existing structures, extend boundaries, or occupy regions that preserve compositional balance.

8.2 Developmental stages as metric refinement

Goodnow’s progression from simple attachment rules in early childhood to symmetric, hierarchically organized compositions in older children can be interpreted as the progressive acquisition of a more refined metric on compositional state space. Young children use a coarse metric dominated by proximity; older children weight symmetry, distribution, and relational coherence alongside proximity.

8.3 Kellogg’s primitive trajectories

Rhoda Kellogg’s survey of children’s early drawing identifies recurring geometric primitives [10]: loops, spirals, zigzags, arches, and radiating lines. These correspond to the simplest geodesic shapes in the plane (constant-curvature curves, piecewise-linear paths, self-similar expanding paths). Their cross-cultural universality supports the claim that the trajectory grammar is prior to, and independent of, semantic representation—children explore the geometry of gesture before they represent objects.

Viktor Lowenfeld’s developmental stages from scribble through schematic to realistic representation [12] map cleanly onto this progression: the scribble stage is kinesthetic trajectory exploration; the schematic stage is the emergence of stable object-templates as trajectory prototypes; realistic drawing is the beginning of field-based representation.

9 QWERTY Swipe as a Trajectory Language

9.1 The keyboard as an embedded graph

Let $K = (V, E)$ be the QWERTY keyboard graph with vertex coordinates $\phi : V \rightarrow \mathbb{R}^2$. This makes K a finite metric space in the induced Euclidean metric.

Definition 9.1 (Swipe trajectory). For a word $w = w_1 \cdots w_n \in \Sigma^*$, the *swipe trajectory* is the piecewise-linear path $P(w) = (\phi(w_1), \phi(w_2), \dots, \phi(w_n)) \subset \mathbb{R}^2$.

Theorem 9.2 (Trajectory encoding). *The map $P : \Sigma^* \rightarrow \mathcal{T}$ is a nontrivial homomorphism from the free monoid $(\Sigma^*, \cdot)$ into the monoid $(\mathcal{T}, \cdot)$ of piecewise-linear planar paths under concatenation. Hence QWERTY swipe induces a formal trajectory language.*

Proof. Concatenation $w = uv$ maps to path concatenation $P(w) = P(u) \cdot P(v)$ (joined at $P(u)$'s endpoint), giving the monoid homomorphism property. Nontriviality holds because distinct words generically produce distinct polygonal chains. $\square$

9.2 Geometric invariants

Each word w has a geometric signature:

$$\begin{aligned} L(w) &= \sum_{i=1}^{n-1} \|\phi(w_{i+1}) - \phi(w_i)\| && \text{(path length),} \\ C(w) &= \sum_{i=2}^{n-1} |\theta_i| && \text{(turn complexity),} \\ X(w) &= \text{(self-intersection count),} \\ \kappa_{\text{close}}(w) &= 1 - \frac{\|\phi(w_n) - \phi(w_1)\|}{L(w)} && \text{(closure index).} \end{aligned}$$

Definition 9.3 (Trajectory equivalence). Words $w, u \in \Sigma^*$ are *trajectory-equivalent* if $d_F(\tilde{P}(w), \tilde{P}(u)) < \varepsilon$, where $\tilde{P}$ is the arc-length-normalized version of P and d_F is Fréchet distance.

9.3 Geometric taxonomy of the lexicon

Shape class	Example	Dominant invariant
Loop / circle	<i>tent</i>	high κ_{close}
Rectilinear	<i>route</i>	few turn directions
Braided	<i>magic</i>	$X(w) \geq 1$
Arch	<i>con</i>	smooth convex arc
Hook	<i>do</i>	large single turn, short L
Ribbon sweep	<i>all</i>	large L , small C
Spiral	<i>crying</i>	increasing radius
Zigzag	<i>Zeno</i>	alternating diagonal segments

Corollary 9.4. *Any lexicon $\mathcal{L} \subset \Sigma^*$ inherits a trajectory semantics via P , making it simultaneously a symbolic language and a geometric codebook. The natural-language vocabulary occupies a proper submanifold of the full swipe trajectory space.*

The swipe keyboard thus corresponds to the painter’s canvas in discretized form: instead of depositing pigment continuously, the gesture snaps to discrete letter centers. The dual reading of a swipe word is $w \mapsto (\text{semantic meaning}, \text{gesture class})$. This was first formalized computationally in the SHARK² system of Kristensson and Zhai [11], which demonstrated that large-vocabulary shorthand writing can be recovered from gesture trajectories alone.

10 Topology of Gesture Classes

Trajectory languages can be analyzed through topological invariants preserved under continuous deformation without crossing a forbidden region.

Definition 10.1 (Topological equivalence). Two trajectories $\gamma_1, \gamma_2 \in \mathcal{T}$ are *topologically equivalent* if one can be continuously deformed into the other within $\mathbb{R}^2 \setminus \mathcal{O}$, where $\mathcal{O}$ is the set of obstacle regions (keyboard edges, object boundaries).

Key invariants include:

$$\begin{aligned} X(\gamma) &= \text{signed self-intersection count,} \\ \omega(\gamma) &= \text{winding number around a reference point,} \\ \kappa_{\text{sign}}(\gamma) &= \text{signed curvature sequence.} \end{aligned}$$

These partition trajectory space into gesture classes:

Class	Topological invariant	Example
Loop	$\omega = 1$	circular swipe or closed brushstroke
Spiral	$\omega > 1$	expanding trajectory
Zigzag	alternating curvature sign	segmented gesture
Pretzel	$X \geq 1$	self-crossing trajectory

The significance is that human motor systems naturally produce gestures that remain stable within a topological class: small variations in speed or curvature do not change the qualitative shape. Swipe keyboards tolerate large motor variability precisely because recognition operates on topological class membership rather than exact geometric matching. This also explains why artists develop characteristic stroke families that constitute a personal topological vocabulary of movement.

11 Eye Saccades and Embodied Perception

11.1 Saccades as perceptual sampling trajectories

Visual perception does not occur through smooth continuous scanning. The eye moves through sequences of rapid fixation jumps called *saccades*. If fixation points are $p_1, p_2, \dots, p_n \in \mathcal{I}$, the visual scan path is a trajectory $\gamma_{\text{eye}}(t)$ through the image plane.

Yarbus [16] showed that scan paths are not random: they are governed by a *saliency field* $\sigma : \mathcal{I} \rightarrow \mathbb{R}_{\geq 0}$, where high saliency corresponds to edges, faces, motion, and high-contrast regions. The saccade trajectory approximately maximizes accumulated saliency subject to a motor cost for large eye movements—a constrained optimization structurally identical to the variational principle of Theorem 4.1.

11.2 Three trajectory systems

Three trajectory systems operate jointly in visual culture:

System	Body part	Function
Eye saccades	Ocular muscles	Perceptual sampling
Brushstrokes	Arm and hand	Image construction
Swipe gestures	Finger	Symbol generation
All three: $\gamma(t)$ through a structured field.		

The hand trajectory of a painter tends to follow the same image features that the eye’s saccades follow (edges, contours, saliency peaks), because both are guided by the same gradient field. The painter is externalizing the visual scan path as a stroke.

11.3 Mimetic hypothesis and motor simulation

Cox’s *mimetic hypothesis* holds that music perception recruits internal motor representations [6]. The perceptual trajectory in that case moves through *motor activation space*. Central pattern generators (CPGs) provide a neural substrate: a CPG chain produces rhythmic sequences of motor states $m_{t+1} = F(m_t)$, a trajectory through motor state space driveable by auditory input. Baddeley’s phonological loop [3] is a cyclic CPG:

$$a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_n \rightarrow a_1,$$

maintaining verbal information through repeated traversal of an articulatory trajectory.

The unified claim is:

Perception and expression across modalities share a common structure: they are trajectories through structured fields, whether those fields are spatial gradients on a canvas, geometric lattices on a keyboard, saliency maps in an image, or motor activation manifolds in the nervous system.

12 Trajectory Space and the Painter’s Trace

12.1 Trajectories as field-modifying traces

The painter’s gesture $\gamma_{\text{paint}} : [0, T] \rightarrow \Omega$ deposits pigment along its trace $\text{Tr}(\gamma) = \{\gamma(t) : t \in [0, T]\}$. Unlike the cinematic case, where γ merely samples a pre-existing $\mathcal{S}$, a brushstroke actively modifies the field it moves through: each stroke changes the canvas over which subsequent strokes will be laid. This creates a dynamic feedback between trajectory and environment absent from cinematography.

The equivalence of trajectory types is now complete:

System	Field	Trajectory product
Swipe typing	keyboard graph K	symbolic word
Brushstroke	canvas B_t (evolving)	pigment distribution
Eye saccade	saliency map σ	sampled percept
Camera path	scenario manifold $\mathcal{S}$	rendered film
All four: $\gamma(t)$ through a field with guidance vector F .		

12.2 Painting as a continuous gesture language

Painting may be understood as a continuous gesture language whose vocabulary consists of trajectory primitives. Just as words correspond to characteristic swipe shapes on a keyboard, painterly marks correspond to characteristic trajectory families in the plane. The geometric primitives of Kellogg’s catalog (loops, spirals, arches, zigzags, hooks) appear both in early children’s drawings and in the gestural marks of mature painters, arising because they are stable trajectories under simple vector fields.

12.3 Perception as trajectory reconstruction

When a viewer observes a painting, the eye’s saccadic path tends to follow the same edges and contours that guided the painter’s hand. Perception retraces the trajectories that produced the image. The painting is not merely an object but a *frozen trajectory system*: its geometry encodes the gestures that created it, and perception partially reconstructs those gestures.

13 Curvature Control in Cinematic Motion

Beyond geodesic smoothness, cinematic trajectories exhibit characteristic curvature profiles that influence the viewer’s perception of motion.

Definition 13.1 (Perceptual curvature). Let $\tilde{\gamma}(t) = \Pi(\gamma(t))$ be the perceptual trajectory. Its *perceptual curvature* at time t is

$$\kappa(t) = \frac{\|\dot{\tilde{\gamma}}(t) \times \ddot{\tilde{\gamma}}(t)\|}{\|\dot{\tilde{\gamma}}(t)\|^3}.$$

Low perceptual curvature corresponds to smooth, drifting motion typical of contemplative cinema; high curvature corresponds to abrupt shifts in experiential orientation. Different cinematic techniques produce characteristic profiles:

Technique	Curvature pattern
Tracking shot	$\kappa(t)$ small and slowly varying
Whip pan	brief spike in $\kappa(t)$
Match cut	curvature discontinuity at the cut boundary
Montage	repeated high- κ transitions
Dolly zoom	κ spike with compensating V -gradient

Proposition 13.2. *The aesthetic smoothness of a camera move correlates with bounded perceptual curvature $\kappa(t)$. Mechanical stabilizers (steadicams, gimbals) produce visually pleasing motion because they implicitly constrain κ in physical space, which bounds κ in $\mathcal{P}$ through the Lipschitz continuity of Π .*

In virtual cinematography the same principle guides camera planning: minimizing $\int_0^T \kappa(t) dt$ in $\mathcal{P}$ rather than in physical space generates movements that feel perceptually natural even when the underlying path in $\mathcal{S}$ is geometrically complex.

14 Developmental Arc: From Object Grammar to Radiance Model

Artistic development can be interpreted as a progressive shift in the representational model underlying trajectory production, moving through three conceptual layers.

Stage 1: Object-oriented. The page is a set of discrete entities. A drawing is an object graph $G = (R, E)$ where regions R_i correspond to symbolic shapes (circle for head, lines for limbs) and edges E encode attachment or containment. Strokes follow object boundaries. This is the stage described by Goodnow’s grammar of action in early childhood.

Stage 2: Field-based. The page becomes a tonal field $I(x, y)$. Strokes follow the gradient and isophote structure of the field rather than symbolic object boundaries. The artist observes continuous phenomena—shadow, reflection, graduated tone—and reconstructs them as trajectories through the gradient field.

Stage 3: Radiance model. The experienced artist understands the image as the projection of a physical light-transport process, implicitly solving the inverse problem encoded by the rendering equation:

$$L_o(x, \omega) = L_e(x, \omega) + \int_{\Omega} f_r(x, \omega', \omega) L_i(x, \omega') (\omega' \cdot n) d\omega'.$$

The artist does not compute this integral explicitly but has internalized how light behaves and places strokes that reproduce its effects.

This three-stage arc corresponds to a progression from a symbolic object metric to an information-geometric field metric as the criterion for naturalness of strokes—exactly the progression the present framework formalizes.

15 Learning Cinematic Trajectories

The variational framework suggests a principled approach to learning cinematic style from observed films.

Given a corpus with associated perceptual state sequences $\tilde{\gamma}_1, \dots, \tilde{\gamma}_N$ (approximated via automatic annotation of attention, composition, and narrative labels), one can learn:

- (a) the metric g_θ on $\mathcal{P}$ by fitting a Riemannian metric to minimize geodesic deviation of observed trajectories;
- (b) the potential V_ϕ by fitting the forced geodesic equation to observed perceptual paths;
- (c) the map Π_ψ by learning to predict perceptual coordinates from scenario states.

A generative cinematography system samples new scenario trajectories $\gamma(t)$ that induce perceptual trajectories $\tilde{\gamma}(t)$ minimizing $\mathcal{J}[\tilde{\gamma}]$ for a specified narrative potential.

Remark 15.1. This framework is compatible with diffusion-based video generation models, where the denoising trajectory can be interpreted as navigation through scenario space. Conditioning on a perceptual metric and narrative potential constrains the generative trajectory to cinematographically natural paths, providing an interpretable inductive bias absent from purely data-driven approaches.

Applications include: automatic cinematography for virtual production; camera planning in game engines; film style analysis and director fingerprinting; trajectory-based video generation with aesthetic constraints; and gesture-driven interfaces for creative tools.

16 Conclusion

We have proposed a unified geometric framework in which diverse cultural practices—cinema, painting, drawing, swipe-gesture input, and visual perception—share a common mathematical structure: directed motion through a structured field, where naturalness is determined by a variational principle on a Riemannian manifold.

The central theoretical contribution is the Geodesic Principle of Cinematic Naturalness (Theorem 4.1): natural camera movement approximates a geodesic in the perceptual manifold $\mathcal{P}$, subject to forcing by a narrative potential V . This principle is grounded in the Fisher-Rao metric on the statistical manifold of viewer attentional distributions (Theorem 6.1), and gives a precise measure-theoretic account of continuity editing as optimal attentional transport. Classical editing conventions emerge as geometric constraints in $\mathcal{P}$: the 180-degree rule is a chart condition (Proposition 5.1); match cuts minimize perceptual distance; montage deliberately exploits high-curvature excursions along semantically productive axes.

The same variational structure governs the painter’s brush (guided by tonal gradient fields and the structure tensor), the child’s drawing hand (constrained by a grammar of spatial action that constitutes a form of metric learning), the typist’s swiping finger (moving through a discrete keyboard manifold that functions as a trajectory codebook, as shown in Theorem 9.2), and the viewer’s eye (whose saccadic path retraces the gradient trajectories that produced the image). Topological invariants of gesture classes provide an additional layer of structure explaining the robustness of gesture recognition to motor variability.

Across all these modalities, meaning and aesthetic quality emerge from the shape of motion through a structured field, not from the endpoint of that motion.

The deeper implication is that human creative and perceptual culture is, to a significant degree, a civilization of gesture: the organized motion of bodies through structured fields, leaving traces that carry meaning because of the geometry of how they were made. A comprehensive understanding of that geometry requires the tools of Riemannian geometry, information geometry, optimal transport, and variational calculus—all of which the present framework brings to bear on the study of human expression.

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